Duration: 110 minutes (Written to take approximately 90 minutes)
Total points available in class: 150, Total points available in take-home: 50. Instructions: Answer all questions with only the use of your mind, a writing utensil, and a single handwritten 3"x5" note card. Partial credit will be awarded for correct work unless otherwise noted.
You must SHOW ALL WORK to receive full credit.
NO Cell phones. Your phone must be in your bag, and out of reach. If it is out during an exam - your exam is compromised and will be taken by the proctor.

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Problem	Score
1 *(10 pts)*	7
2 *(10 pts)*	8
3 *(15 pts)*	15
4 *(15 pts)*	11
5 *(10 pts)*	10
6 *(20 pts)*	20
7 *(15 pts)*	8
8 *(20 pts)*	13
9 *(5 pts)*	3
10 *(5 pts)*	5
11 *(5 pts)*	5
TOTAL ~~*(150 pts)*~~ 130	105

→ 122/180

$ay'' + by' + cy = F(t)$

1. When solving a forced oscillator problem with damping using a Fourier series, how does resonance appear in the solution? Explain in detail.

When resonance appears in a forced oscillator function, it appears with a large amplitude since it's experiencing damping. The resonance appears in the form of a forcing function and the resonance grows unbounded. The dominant mode may be calculated by $\sqrt{c/a}$ in the above eq'n

2. Explain and compare uniform convergence with pointwise convergence.

Pointwise convergence contains a finite # of discontinuities with the limit existing @ each side. Uniform convergence occurs if the periodic function is continuous. In short, both uniform and pointwise ~~test~~ convergence, but under different circumstances.

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not a test, but a type of convergence

3. Classify the following PDEs as *hyperbolic*, *parabolic*, or *elliptic*. Show all work leading to your conclusion. $Au_{xx} + 2Bu_{xy} + Cu_{yy} + Du_x + Eu_y + Fu = G$ / $AC - B^2$

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(a) $u_x - u_{yy} = 0$

(b) $u_{xx} - u_{yy} = 0$

(c) $u_{xx} - 2u_{xy} + u_{yy} = 0$

a.) $A = 0$, $B = 0$, $C = -1$

$AC - B^2 = (0)(-1) - (0)^2$

$= 0 \rightarrow$ Parabolic

b.) $A = 1$, $B = 0$, $C = -1$

$AC - B^2 = (1)(-1) - (0)^2$

$= -1 \rightarrow$ Hyperbolic

c.) $A = 1, B = -1, C = 1$

$AC - B^2 = (1)(1) - (-1)^2$

$= 1 - 1 = 0$

$\rightarrow$ Parabolic

4. Classify the following side conditions (type, homogeneity, and name - if relevant)

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(a) $u(x, 0) = f(x)$

(b) $u(0, t) = a$

(c) $u_x(0, t) = 0$

a.) ~~Dirichlet~~

Non-Homogeneous

P.D.E.

0th order (no derivatives w.r.t. x or t)

b.) ~~N/A (no name)~~

Non-homogeneous

PDE

0th order (no derivatives w.r.t. x or t).

Dirichlet BC

c.) Neumann BC

homogeneous

PDE

First order w.r.t. x derivative of function

5. Separate the following PDE in to a system of ODEs: $u_{xx} - xu_y = 0$ (Do **not** solve - just set up the ODEs.)

$u(x,y) = X(x)Y(y)$

$u_x = X'(x)Y(y)$ $\quad u_y = X(x)Y'(y)$

$u_{xx} = X''(x)Y(y)$ $\quad u_{yy} = X(x)Y''(y)$

$u_{xx} - xu_y = 0 \rightarrow [X''(x)Y(y)] - x[X(x)Y'(y)] = 0$

$X''(x)Y(y) = x \cdot X(x)Y'(y)$

$$\frac{X''(x)}{x \cdot X(x)} = \frac{Y'(y)}{Y(y)} = \lambda$$

$\boxed{X\text{-world}}$

$$\frac{X''(x)}{x \cdot X(x)} = \lambda$$

$X''(x) = \lambda \cdot x \cdot X(x)$

$X''(x) - \lambda \cdot x \cdot X(x) = 0$

✓

$\boxed{Y\text{-world}}$

$$\frac{Y'(y)}{Y(y)} = \lambda$$

$Y'(y) = \lambda Y(y)$

$Y'(y) - \lambda Y(y) = 0$

✓

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6. Solve the eigenvalue problem $y'' + \lambda y = 0$, given the boundary conditions $y(0) = y'(\pi) = 0$ Identify relevant eigenvalues and their associated eigenfunctions (Show all cases leading to your solution).

$y'' + \lambda y = 0$

$\lambda y = -y''$

$y = \frac{-y''}{\lambda}$

$r^2 + \lambda = 0$

$r^2 = -\lambda$

$r = \sqrt{-\lambda} = 0 \pm \lambda i \quad \alpha = 0, \beta = 1\lambda$

$y(x) = C_1 \sin(\sqrt{-\lambda}x) + C_2 \cos(\sqrt{-\lambda}x) \,/\, C_1 \sin(x) + C_2 \cos(x)$

$y'(x) = C_1\sqrt{-\lambda}\cos(\sqrt{-\lambda}x) - C_2\sqrt{-\lambda}\sin(\sqrt{-\lambda}x) \,/\, C_1\cos(x) - C_2\sin(x)$

$y(0) = 0 \rightarrow C_1 \sin(0) + C_2\cos(0) = 0 \,/\, C_2 \cdot 1 = 0 \rightarrow C_2 = 0$

$C_2 = 0$

$y = C_1 \sin(x)$

$y' = C_1 \cos(x)$

$y'(\pi) = 0 \rightarrow 0 = C_1 \cos(\pi)$

$y'(x) = C_1\sqrt{-\lambda}\cos(\sqrt{-\lambda}x)$

$y'(\pi) = 0 \rightarrow C_1\sqrt{-\lambda}\cos(\sqrt{-\lambda}\pi) = 0$

$\cos(\sqrt{-\lambda}\pi) = 0$

$\sqrt{-\lambda}\,\pi = \frac{(2n-1)\pi}{2}$

$\sqrt{-\lambda} = \frac{(2n-1)}{2}$ ✓

$\lambda = \frac{-(2n-1)^2}{4}$

$y_n(x) = C_1 \sin\left(\frac{(2n-1)\cdot x}{2}\right)$

✓ $y(x) = \sum_{n=1}^{\infty} C_n \sin\left(\frac{(2n-1)x}{2}\right)$

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7. Use D'Alembert's solution for the wave equation with homogeneous Dirichlet conditions to determine the solution to the wave equation $u_{tt} = u_{xx}$ on $0 < x < \pi$ with initial conditions $u(x,0) = \sin x$ and $u_t(x,0) = 1$.

$c=1$

$$u(x,t) = \frac{1}{2}\left[f^*(x-ct) + f^*(x+ct)\right] + \frac{1}{2c}\int_{x-ct}^{x+ct} g^*(s)\,ds$$

find the odd extension of 1.

$$u(x,t) = \frac{1}{2}\left[\sin(x-t) + \sin(x+t)\right] + \frac{1}{2}\int_{x-t}^{x+t} \cancel{1}\,ds$$

$$= \frac{1}{2}\sin(x-t) + \frac{1}{2}\sin(x+t) + \frac{1}{2}\{s\}\Big|_{x-t}^{x+t}$$

$$= \frac{1}{2}\sin(x-t) + \frac{1}{2}\sin(x+t) + \frac{1}{2}\{[x+t]-[x-t]\}$$

$$= \frac{1}{2}\sin(x-t) + \frac{1}{2}\sin(x+t) \cancel{+t}$$

(I don't remember the trig identities; please don't dock me.)

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$L = 6CM$

$u_{tt} = c^2 u_{xx}$

$c = 2$

8. Solve $u_{tt} = 4u_{xx}$ for a string of length $6cm$ with BCs $u(0,t) = 0$, $u(6,t) = 6$, and initial conditions $u(x,0) = 5x$ and $u_t(x,0) = 0$. (Note: You CAN assume the form of the series from the problem - you do not need to show all the steps that lead to it for this timed exam.)

$u(x,t) = X(x)T(t)$

$u_t = X(x)T'(t)$

$u_{tt} = X(x)T''(t)$

$u(x,t) = \frac{1}{2}[f^*(x-ct) + f^*(x+ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} g^*(s)\,ds$

$X(x)T''(t) = 4X''(x)T(t)$

$u_x = X'(x)T(t)$

$u_{xx} = X''(x)T(t)$

$\frac{T''(t)}{4T(t)} = \frac{X''(x)}{X(x)} = \lambda$

$T''(t) - 4\lambda T(t) = 0$

$X''(x) - \lambda X(x) = 0$

1.) $\lambda > 0$ $X(x) = c_1 e^{\sqrt{\lambda}x} + c_2 e^{-\sqrt{\lambda}x}$

$X(0) = 0 \to c_1 + c_2 = 0 / c_1 = -c_2$

$X(x) = c_1 e^{\sqrt{\lambda}x} - c_1 e^{-\sqrt{\lambda}x}$

$= 2c_1 \sinh(\sqrt{\lambda}x)$

$X(6) = 2c_1 \sinh(6\sqrt{\lambda}) = 6$

$c_1 = \frac{3}{\sinh(6\sqrt{\lambda})}$

$X(x) = \frac{6\sinh(\sqrt{\lambda}x)}{\sinh(6\sqrt{\lambda})}$

not needed

2.) $\lambda = 0$ $X(x) = \alpha x + \beta$

$X(0) = 0 \to 0 = \beta \to X(x) = \alpha x$

$X(6) = 6 \to \alpha 6 = 6 \to \alpha = 1 \to X(x) = x$ $\to u(x,t)$ [illegible]

3.) $\lambda < 0$ $X(x) = c_1 \sin(\sqrt{-\lambda}x) + c_2 \cos(\sqrt{-\lambda}x)$

$X(0) = 0 \to 0 = c_2 \to X(x) = c_1 \sin(\sqrt{-\lambda}x)$

$X(6) = 6 \to 6 = c_1 \sin(\sqrt{-\lambda}6)$

$u(x,t) = \frac{1}{2}\left[\cancel{5(x-0t)} + 5(x-0t)\right] + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{6}\right)$

$+ \frac{1}{4}\int_{x-ct}^{x+ct} 0\,ds$

$u(x,t) = \cancel{5x} + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{6}\right)$

$b_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)dx$

$= \frac{1}{3}\int_0^6 5x \sin\left(\frac{n\pi x}{6}\right)dx$

$= \frac{5}{3}\int_0^6 x\sin\left(\frac{n\pi x}{6}\right)dx = \frac{-6x}{n\pi}\cos\left(\frac{n\pi x}{6}\right) + \frac{36}{n^2\pi^2}\sin\left(\frac{n\pi x}{6}\right)\Big|_0^6$

$= \left\{\left[\frac{-36}{n\pi}(-1)^n + 0\right] - [0 + 0]\right\}$

$= \frac{-36(-1)^n}{n\pi}$

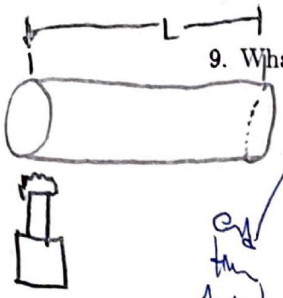
$+x$	$\sin\left(\frac{n\pi x}{6}\right)$	
-1	$\frac{-6}{n\pi}\cos\left(\frac{n\pi x}{6}\right)$	
$+0$	$\frac{-36}{n^2\pi^2}\sin\left(\frac{n\pi x}{6}\right)$	

$u(x,t) = \cancel{5x} + \sum_{n=1}^{\infty} \frac{-36(-1)^n}{n\pi}\sin\left(\frac{n\pi x}{6}\right)$

$\cancel{\frac{1}{4}\int_{x-ct}^{x+ct} 0\,ds} = \frac{1}{4}\left\{[x+ct] - [x-ct]\right\} = \frac{2ct}{4} = \frac{ct}{2}$

$= \cancel{t}$

13 $u(x,t) = \cancel{5x + t} + \sum_{n=1}^{\infty} \frac{-\cancel{36}^{6}(-1)^n}{\pi n}\sin\left(\frac{n\pi x}{6}\right)$



9. What is the steady-state temperature distribution in the Heat equation? Explain.

Steady-state temp. distri. in the heat eq'n means that a rod of length L will experience an initial, uniform heat/temp. distribution throughout the rod before undergoing a change in temperature. For example if a blow torch were to heat up a rod, the rod would, initially, have a temperature before experiencing a change in temp. to be modeled by the solution. ?

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10. Describe the differences in solutions for the wave equation and the heat equation.

wave: $u_{tt} = c^2 u_{xx}$ heat: $u_t = K u_{xx}$ since we only have to take one derivative of the $u(x,t)$ function w.r.t. t + 2 w.r.t x in the heat eq'n as opposed to two derivatives w.r.t. both t + x in the wave eq'n. This means that wave sol'ns have equal derivative values while heat does not. Also, heat solutions experience an exponential decay represented by the $e^{-(\text{constant} \cdot \text{variable})}$ term while wave does not. Wave eq'ns also have their solutions be strictly waves (go figure) while heat does not.

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11. How were the solutions affected when solving the 2D Wave and Heat equations compared with the 1D versions?

When setting up the equations, we need to use multiple variables in the beginning (i.e. $u_{tt} = c^2(u_{xx} + u_{yy})$ + $u_t = K(u_{xx} + u_{yy})$) and plug in the appropriate $X(x)$, $Y(y)$ + $T(t)$ functions before solving. Our solutions were therefore in the form of additional variables to reflect these extra functions, and also had an additional function (i.e. $u(x,y,t) = F_1(x) \cdot F_2(y) \cdot F_3(t)$) to boot. It was also harder to graph the solutions since they used additional variables.

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